

THIS ISSUE TRAINS · RECOGNISING WHEN A SUBSTITUTE CHANGES THE GAME

Money and markets

Sodium ion batteries made MIT Technology Review's 10 Breakthrough Technologies for 2026. Made from salt and abundant materials instead of lithium, they are emerging as a cheaper, safer alternative for grid storage and affordable electric vehicles. The batteries are not better than lithium ion on every measure, but they do not need cobalt or nickel, which means they sidestep the supply chains and prices that make lithium batteries expensive.

For investors, the question is not whether sodium ion is technically superior. It is whether it is good enough at a low enough price to take market share from lithium. History says good enough and cheap usually wins.

The SCD takeaway: a product does not need to be better than the incumbent. It needs to be good enough and cheaper. That combination has ended more monopolies than any regulation.

Sources: MIT Technology Review, 10 Breakthrough Technologies 2026.

Engineering corner

A battery made from salt. Sodium ion batteries use sodium, one of the most abundant elements on Earth, in place of lithium. They store less energy per kilogram, roughly 30 to 40 percent less, but they are safer (less prone to thermal runaway), tolerate a wider temperature range, and can be manufactured using existing lithium ion production lines with minor modifications. For stationary grid storage, where weight does not matter, the cost advantage outweighs the energy density penalty.

The SCD takeaway: the best material for a job is not always the highest performing one. It is the one whose limitations do not matter for the specific application.

Source: MIT Technology Review, 10 Breakthrough Technologies 2026.

Your move

A substitute does not need to match the original on every dimension. It only needs to win on the dimensions the buyer actually cares about. Resistance to substitutes is often loyalty to a spec sheet rather than to a need.

This week, on your own: find one product you use where a cheaper alternative exists that you have never tried. Write down what exactly stops you from switching. Is it performance, habit, or something else?